

Enhancing Bank Customer Service with Generative AI

Development and Evaluation of a GenAI-Powered Chatbot

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Agenda

Background

Project 1: Customer Service Chatbot

Project 2: Chatbot Evaluation System

Internship Takeaways

Summary

Background

Company Background

- ▶ Singapore-based bank focusing on transforming banking services.
- ▶ Targets everyday consumers and small businesses.
- ▶ Emphasizes advanced technology and data analytics.

Project Background

- ▶ **Customer Expectations:**

- ▶ Demand for seamless and efficient support.
- ▶ Need for immediate access to information in digital banking.

- ▶ **Solution:**

- ▶ Development of a GenAI-powered chatbot utilizing publicly available FAQ articles.
- ▶ Application of RAG to answer user queries with internal data.

- ▶ **Objectives:**

- ▶ Increase self-service rates.
- ▶ Provide quick access to information.
- ▶ Enhance customer support efficiency.

Introduction to Generative AI

▶ **What is Generative AI (GenAI)?**

- ▶ Innovative technology in natural language processing.
- ▶ Capable of producing coherent and contextually relevant text.

▶ **Core Components:**

▶ **Large Language Models (LLMs):**

- ▶ Developed using deep learning and transformer architectures.

▶ **Transformers:**

- ▶ Efficiently capture relationships across words and phrases [16].

▶ **GPT-3 and Beyond:**

- ▶ Demonstrated exceptional few-shot learning capabilities [2].
- ▶ Basis for current GenAI applications.

Risks Associated with GenAI in Banking

▶ **Hallucination:**

- ▶ Models may generate plausible but incorrect responses [8].
- ▶ Potential for misinformation and legal liability.

▶ **Black Box Nature:**

- ▶ Limited transparency in model decision-making [4, 14].
- ▶ Challenges in interpreting and explaining responses.

▶ **Ethical and Security Concerns:**

- ▶ Potential biases in responses [8].
- ▶ Risk of accidental exposure or unauthorized access to personal information [9].
- ▶ Susceptibility to prompt injection attacks [6].

Project 1: Customer Service Chatbot

Project Overview

- ▶ **Objective:**

- ▶ Develop a secure and efficient customer service chatbot.

- ▶ **Challenges Addressed:**

- ▶ Security.
- ▶ Hallucination.
- ▶ Data handling.

- ▶ **Approach:**

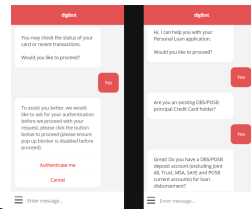
- ▶ Multi-phase and multi-agent methodology.
- ▶ Robust architecture ensuring data privacy and scalability.

Market Solutions

► Existing Implementations:

► DBS Bank's Digibot[3]:

- Utilizes a rule-based system architecture.
- Focuses on operational reliability over conversational flexibility.
- Incorporates security protocols like two-factor authentication and end-to-end encryption.
- Effective for routine operations (loan applications, transaction history, account info).
- Limited semantic understanding, relies on keyword recognition.
- Prioritizes transactional accuracy within defined parameters.



Market Solutions

- ▶ **Bank of America's Erica[1]:**
 - ▶ Employs natural language processing and machine learning, excluding GenAI.
 - ▶ Uses traditional ML models for query interpretation and response selection.
 - ▶ Includes continuous learning mechanisms for response refinement.
 - ▶ Handles standard banking functions and proactive financial monitoring.
 - ▶ Operates within authenticated mobile application environment.



Market Solutions

- ▶ **JPMorgan Chase's IndexGPT[10]:**
 - ▶ Integrates OpenAI's GPT-4 for institutional banking.
 - ▶ Functions in thematic keyword generation for investment portfolio construction.
 - ▶ Enhances index construction process efficiency with human oversight.
 - ▶ Balances advanced AI capability with operational risk management.

Chatbot Interface

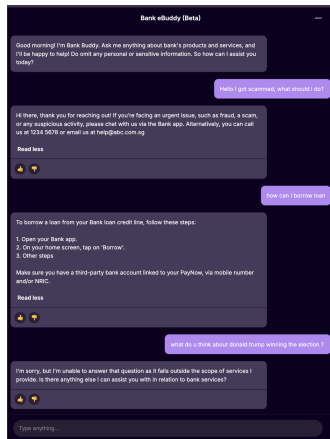


Figure: Frontend Interface of the Chatbot for User Interaction.

Methodology

1. Session Initialization and Data Sanitization

- ▶ Establish user session.
- ▶ Sanitize inputs to remove sensitive data.

2. Language Detection and Greeting Handling

- ▶ Detect input language and verify support.
- ▶ Identify and respond to greetings appropriately.

Methodology

3. Context Management and Security Verification

- ▶ Maintain conversation context.
- ▶ Employ the LLM Security Agent to detect and mitigate threats.

4. FAQ Retrieval and Response Generation

- ▶ Retrieve relevant FAQs using the LLM Retrieval Agent.
- ▶ Generate accurate, context-aware responses.

5. Output Sanitization and Session Logging

- ▶ Sanitize responses to remove Personally Identifiable Information (PII).
- ▶ Securely log conversation history.

Chatbot System Architecture

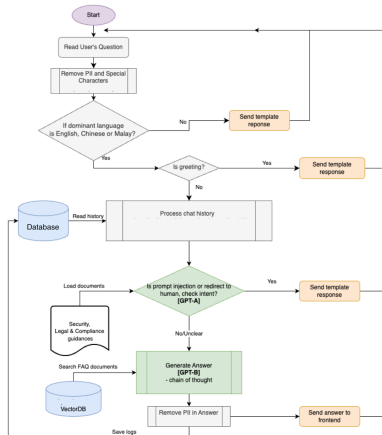


Figure: Flowchart of the Chatbot System Architecture.

Results

▶ **Testing:**

- ▶ Conducted over two months.
- ▶ 2000 user sessions.

▶ **Performance Metrics:**

- ▶ **Latency:** Average response time of 4.5 seconds per interaction.
- ▶ **User Satisfaction:**
 - ▶ 5% positive feedback (thumbs-up).
 - ▶ 4% negative feedback (thumbs-down).
- ▶ **Security Effectiveness:**
 - ▶ 100% detection and mitigation of simulated threats, such as prompt injections and sensitive queries.

Discussion

► Design Objectives:

- Focus on efficiency, accuracy, and data privacy.
- Layered verification and security approach to mitigate risks and safeguard data.

► Strengths:

- Layered verification and security: Content Filtering and PII detection.
- Contextual accuracy with vector-based retrieval aligned with FAQ content.
- Scalability and efficiency with specialized agents (e.g., LLM-A for security checks, LLM-B for retrieval).

► Limitations:

- Noticeable latency due to sequential processing.

Discussion

► **Challenges and Future Improvements:**

- Concurrent phase execution to reduce latency.
- Fine-tuning smaller LLMs for specific tasks.
- Implementing a distributed check system to enhance efficiency.
- Aim to reduce latency and enhance efficiency by running security checks and response generation concurrently.

Project 2: Chatbot Evaluation System

Background

- ▶ **Need for Evaluation:**

- ▶ Continuous monitoring due to LLM limitations.
- ▶ Ensure compliance after model updates or changes.

- ▶ **Challenges:**

- ▶ Manual evaluation is time-consuming and subjective.

- ▶ **Solution:**

- ▶ Development of an automated evaluation system.
- ▶ Utilize a database of 800 curated test questions.

Literature Review

- ▶ **Traditional Metrics:**

- ▶ **ROUGE and BLEU** Focus on textual similarity. Measures n-gram and longest common subsequence overlaps [11, 13].

- ▶ **Advanced Metrics:**

- ▶ **BERTScore:** Uses transformer embeddings to assess semantic similarity [17].
- ▶ **G-Eval and GPTScore:** Evaluates multiple dimensions like accuracy and coherence via LLMs [12, 7].
- ▶ **RAGAS Framework:** Combines model and rule-based approaches for clause extraction and scoring [5].
- ▶ **ARES:** Integrates synthetic data, LLM judgment, and human validation for better alignment [15].

Methodology

- ▶ **Initial Setup:**
 - ▶ Implementation using the RAGAS framework.
- ▶ **Question Base Creation:**
 - ▶ 800 test questions across four domains:
 - ▶ Ethics & Fairness.
 - ▶ Operational Risk.
 - ▶ Exclusion List Compliance.
 - ▶ Product Scenarios.
- ▶ **Ground Truth Creation:**
 - ▶ Responses generated and validated by experts.

Methodology

- ▶ **Metric Selection:**

- ▶ **Retrieval Scores:**

- ▶ Context Recall.
 - ▶ Context Precision.

- ▶ **Generation Scores:**

- ▶ Faithfulness.
 - ▶ Answer Correctness.

- ▶ **Evaluation Pipeline:**

1. Extraction of test questions.
2. Processing through chatbot API.
3. Computation of performance metrics.
4. Post-processing and storage of results.

Methodology

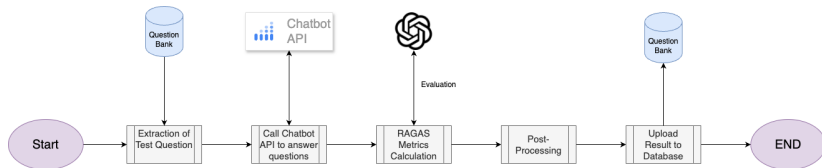


Figure: Flowchart of the Chatbot System Architecture.

Methodology

QUESTION_TYPE	CONTEXT_PRECISION	FAITHFULNESS	CONTEXT_RECALL	ANSWER_CORRECTNESS	NUMBER_OF_QUESTIONS	NUM_OF_TEMPLATE_ANSWERS	TEMPLATE_ANSWER_RATE	RESULT	FAILED_METRICS
additional_qa	0.81	0.77	0.66	0.6	11	0	0%	fail	['faithfulness']
ethics_fairness	0	0	0	0.94	16	14	88%	fail	['answer_correctness']
exclusion_list	0	0	0	1	130	117	90%	pass	[]
focus_group	0.6	0.58	0.7	0.63	34	2	6%	pass	[]
gpt_generated_q	0.7	0.85	0.87	0.7	331	21	6%	pass	[]
internal_beta_lau	0.65	0.74	0.58	0.58	53	1	2%	pass	[]
language	0.75	0.78	0.7	0.73	41	0	0%	pass	[]
multi-language	0.82	0.77	0.65	0.63	55	0	0%	pass	[]
operational_risk	0.82	1	0.77	0.58	9	0	0%	pass	[]
other_exclusion	0	0	0	1	59	25	42%	pass	[]
product_scenario	0.76	0.9	0.67	0.64	56	1	2%	pass	[]

Figure: Example Output from the Chatbot Evaluation System.

Use Cases

▶ **Weekly Monitoring:**

- ▶ The evaluation system runs weekly to monitor chatbot performance.
- ▶ Detects performance drift and triggers alerts for declines.

▶ **Diverse Question Categories:**

- ▶ Assesses chatbot on data privacy, product details, and company regulations.
- ▶ Pinpoints areas of underperformance for targeted improvements.

▶ **Comparison of Chatbot Versions:**

- ▶ Enables systematic comparison of different chatbot versions.
- ▶ Tracks impact of changes such as prompt engineering and model updates.

Discussion

► Advantages of RAGAS Framework:

- Comprehensive evaluation covering multiple pipeline stages for precise performance insights.
- Metrics align with human judgment, mirroring real-world quality assessments.
- Supports custom models, ensuring compliance with guidelines.
- Extensive documentation and active community support (7.1k GitHub stars).
- Zero-shot operation with no need for fine-tuning, yet customizable.

Discussion

Scenario	Exists in Knowledge Base	RAG	Chatbot Answer	Ground Truth	Affected Metric
Information not in knowledge base, but in prompt	no	wrong	Correct	correct	Faithfulness, context_precision, context_recall
The most related info are not placed in the top	yes	wrong	correct	correct	context_precision
Answer is opposite of the ground truth	yes	correct	incorrect	correct	answer_correctness
Incorrect/Outdated info in ground truth	yes	correct	correct	incorrect	context_recall answer_correctness
Correct but missing some details	yes	correct	correct	correct	Context_precision answer_correctness

Figure: Explanation of RAGAS Metric and Its Application Across Different Scenarios.

Discussion

► Limitations of RAGAS Framework:

- Requires predefined questions and ground-truth answers (manual setup).
- Complex metrics that are difficult to interpret comprehensively.
- Evaluation scores can vary due to non-determinism in model responses.

► Mitigation Strategies:

- Use chatbot to generate ground-truth and regularly revise it to maintain quality.
- Focus on four key metrics for clarity and representativeness.
- Average multiple evaluations and set temperature to zero for consistent responses.

Internship Takeaways

Key Learnings

- ▶ **Advanced Coding Practices:**
 - ▶ Emphasis on modularization, reusability, comprehensive documentation, and effective commenting.
- ▶ **GenAI Libraries Proficiency:**
 - ▶ Expertise in libraries such as Lambda Index, LangChain, and OpenAI APIs, with an emphasis on strategic library selection for optimized performance.
 - ▶ Addressing challenges with emerging libraries like RAGAS through source code analysis and independent problem-solving due to limited documentation.
- ▶ **MLOps Infrastructure:**
 - ▶ Hands-on experience with AWS services including Lambda, SageMaker, and Airflow.

Key Learnings

- ▶ Proficient in distributed workflow management, efficient resource allocation, and deploying model evaluation pipelines in high-stakes environments.
- ▶ **Data Privacy and Security:**
 - ▶ Ensured secure handling of PII and internal data in financial services by integrating privacy-preserving mechanisms and adhering to data security standards.
- ▶ **Professional and Communication Skills:**
 - ▶ Effectively conveyed complex AI methodologies to non-technical stakeholders through infographics, PowerPoint, and summaries.
 - ▶ Enhanced technical communication by collaborating with data science and engineering teams.
 - ▶ Strengthened research capabilities through exploration of best practices and industry literature.

Summary

Summary

► **Impact of the Projects:**

- Internship focused on applying GenAI to improve banking customer service.
- Used RAGAS framework for performance monitoring and iterative improvements.

► **Personal Growth:**

- Technical skills in GenAI and data science.
- Banks' data privacy and security measures.
- Importance of team communication and collaboration.

Q&A

Questions and Discussion

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